

Lab 2 Activity

In this lab, we will create some visualizations of the `diamonds` dataset. Use Quarto to create a Word or pdf document of your results. When you are done (or when lab time runs out), check in with the TA to receive participation credit for the day. **This assignment is not graded aside from participation credit**

First make sure that you download the “diamonds.csv” dataset. Make sure that this file is in your working directory (you may want to change your working directory). See Lab 1 for help with this step.

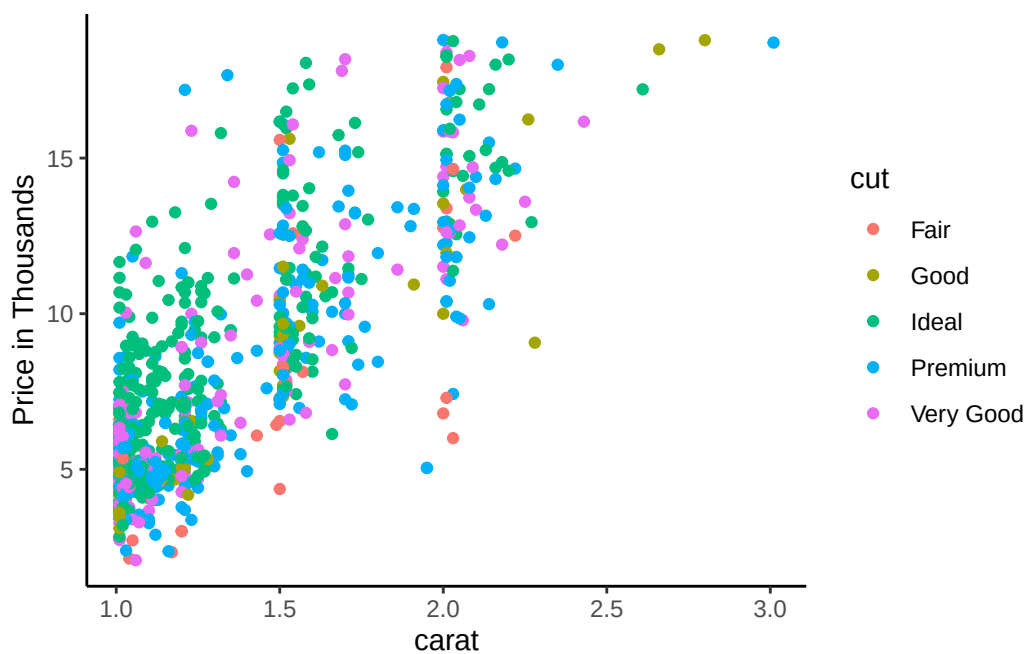
```
library(tidyverse)
diamonds <- read.csv("diamonds.csv")
```

You can run the `?diamonds` command in your console to get a description of the variables in the dataset. Additionally, check out [this page](#) for even more diamond knowledge!

Recreate the following graphs to the best of your ability (feel free to start with any one you please). Use the `ggplot` help pages and the internet to help you! We don’t expect you reproduce every aspect of every plot today, but we want to give you some real-world problem-solving practice. After lab today, we’ll give you the source code for you to review on your own time.

Plot 1

```
diamonds %>%  
  filter(carat > 1) %>%  
  mutate(price_thousands = price/1e3) %>%  
  ggplot(aes(x = carat,  
             y = price_thousands,  
             color = cut)) +  
  geom_point() +  
  labs(y = "Price in Thousands") +  
  theme_classic()
```

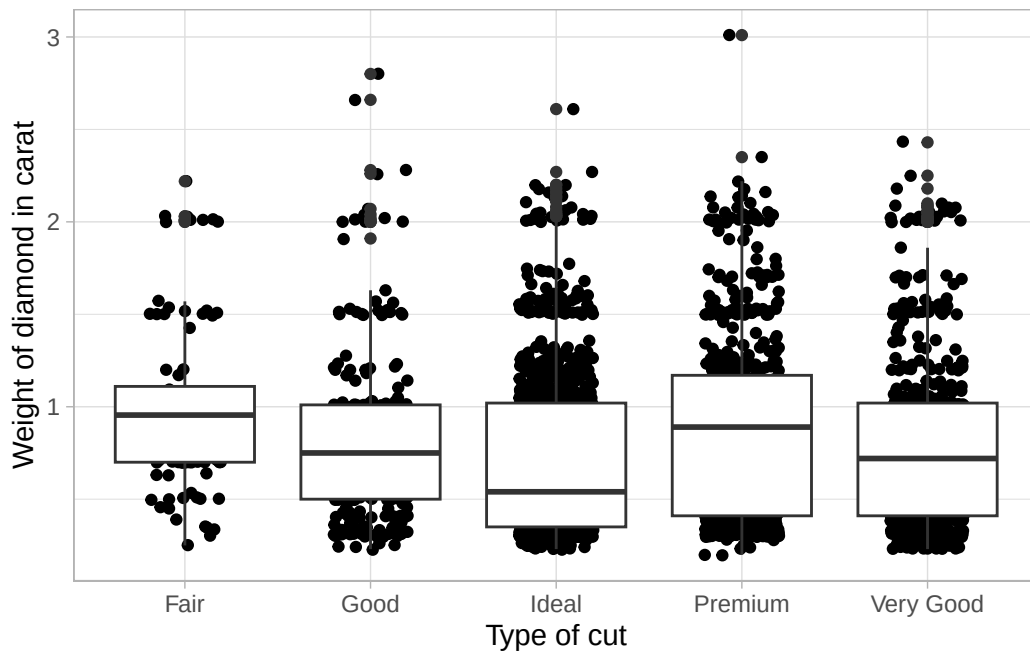


Plot 1 checklist:

- Filter only diamonds that are above 1 carat
- Transform **price** (in USD) to thousands of USD
- Plot different cuts in different colors
- Create a scatterplot with **price** on x-axis and **carat** on y-axis
- Make sure y axis label matches the plot
- Choose a plotting theme that omits the background grid lines

Plot 2

```
diamonds %>%  
  ggplot(aes(x = factor(cut), y = carat)) +  
  geom_jitter(width = .2) +  
  geom_boxplot() +  
  labs(x = "Type of cut",  
       y = "Weight of diamond in carat") +  
  theme_light()
```

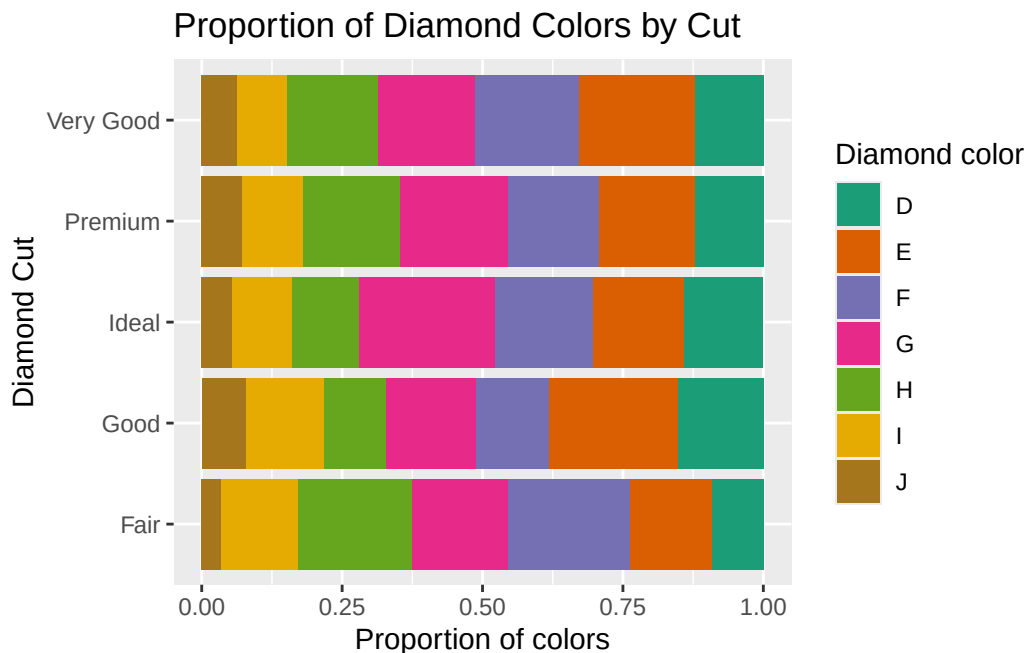


Plot 2 checklist

- cut should be converted to a factor type variable
- Create boxplots
- Create jittered scatterplots with width of .2
- Choose a theme that matches the picture
- Match x and y axis labels

Plot 3

```
diamonds %>%
  group_by(cut, color) %>%
  summarise(color_per_cut = n()) %>%
  group_by(cut) %>%
  mutate(proportion_color = color_per_cut/sum(color_per_cut)) %>%
ggplot(aes(x = cut,
           y = proportion_color,
           fill = color)) +
coord_flip() +
geom_bar(stat = "identity", position = "fill") +
labs(
  title = "Proportion of Diamond Colors by Cut",
  x = "Diamond Cut",
  y = "Proportion of colors",
  fill = "Diamond color") +
scale_fill_brewer(palette = "Dark2")
```



Plot 3 checklist:

- For each cut, calculate the proportion of each colors. The `group_by()` and `summarise()` functions should be quite helpful to do this.

- Create a stacked bar plot that fills the plotting space
- Use different colors for different diamond colors
- Different cuts should be on different rows
- Match the axis and the title
- To match colors exactly, use the “Dark2” palette from the `RColorBrewer` package (see `scale_fill_brewer()` function in `ggplot2`)